

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

4th Semester of B. Tech. Examination (EE)

May 2014

EE205 ELECTRICAL MACHINES - I

Date: 03.05.2014, Saturday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Explain singly excited magnetic field system and also define the terms energy and co-energy. [07]
- Q - 2 (a) Explain parallel operation of DC shunt generators. [06]
- (b) Two DC generators are connected in parallel to supply a load of 1500 A. One generator has an armature resistance of 0.5Ω and emf of 400V. While other has an armature resistance of 0.04Ω and an emf of 440V. The resistances of shunt field are 100Ω and 30Ω respectively. Calculate the current I_1 and I_2 supplied by individual generator and terminal voltage of the combination. [08]

OR

- Q - 2 (a) Explain armature reaction and calculate the demagnetizing AT per pole & cross magnetizing armature AT per pole. [07]
- (b) The armature of a 4 pole DC shunt generator is lap wound and generates 216V. When running at 600 rpm, the armature has 144 slots with 6 conductors per slot. If the armature wound wave connected. Find the emf generated with the same flux per pole but running at 500rpm. [07]
- Q - 3 (a) Explain necessity of starter in DC motor and write short note on three point starter. [08]
- (b) Determine developed torque and shaft torque of 220V, 4 pole series motor with 800 conductors wave connected supplying a load of 8.2 KW by taking 45A from the mains. The flux per pole is 25mWb and its armature circuit resistance is 0.6Ω . [06]

OR

- Q - 3 (a) Explain methods for speed control of DC shunt motor and DC series motor. [08]
- (b) Explain characteristics of DC shunt and compound motor. [06]

SECTION - II

- Q - 4 (a) Answer the following questions. [07]
- (1) Give the names of methods of cooling of transformer.
 - (2) Define parallel operation of transformers.
 - (3) What is the purpose of open circuit test and short circuit test on transformer?
 - (4) Define voltage regulation of transformer and write the equation.
 - (5) Write emf equation of transformer and explain each term.

- (6) Explain the term magnetostriction in brief.
 (7) Classify vector groups based on phase displacement.

- Q - 5 (a) Derive equivalent circuit of transformer with necessary diagrams. [07]
 (b) Find the all day efficiency of 500 KVA distribution transformer whose copper loss and Iron loss at full load are 4.5 KW and 3.5 KW respectively. Load duration is given below [07]

No of hours	Loading in KW	Power factor
6	400	0.8
6	300	0.75
8	100	0.8
4	0	-

OR

- Q - 5 (a) Explain auto transformer and derive equation of saving of copper in auto transformer. [07]
 (b) 1 KVA , 200V/100V, 50Hz single phase transformer gives following test readings [07]
 O.C. Test : $V=100V$, $I_0=0.24A$, $W_i=15W$
 S.C. Test : $V_{sc}=11.5V$, $I_{sc}=7.4A$, $W_{sc}=70W$.
 Calculate circuit parameters, efficiency and voltage regulation at half load and unity power factor.

- Q - 6 (a) Draw vector diagram and phasor diagram for phase group connections Yy_0 , Dd_6 and Yz_{11} . [07]
 (b) Explain tertiary winding in context with its purpose, operating principle, construction and equivalent circuit. [07]

OR

- Q - 6 (a) Explain open delta connection of transformer in detail with example. [07]
 (b) Write short note on initial rush current of transformer. [07]
